



Documentation

Welcome to the documentation for the UE5 City Environment!

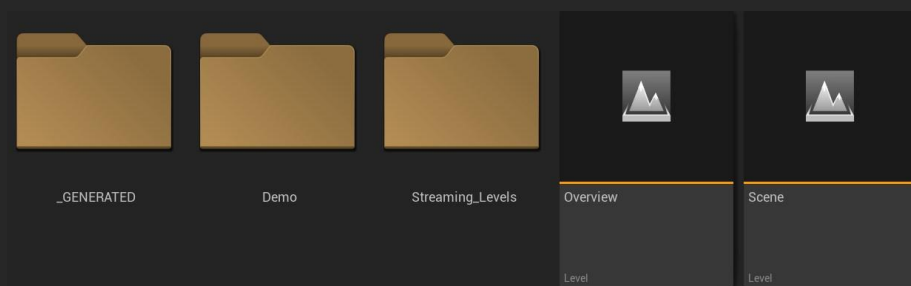
We will keep this documentation short and powerful, in this documentation you can find information on how the environment is constructed, how to control the levels and turn on/off sections of the environment and other general information.

Support email: support@scansmatter.com

Getting Started

*This project uses **Lumen**, you can learn how to activate it [HERE](#) if needed.*

When loading up the project for the first time it should open on the city level right away. If not then you can navigate to “City_Environment> Scene” Where you can find multiple levels. (please note first time might take a while due to the map size)

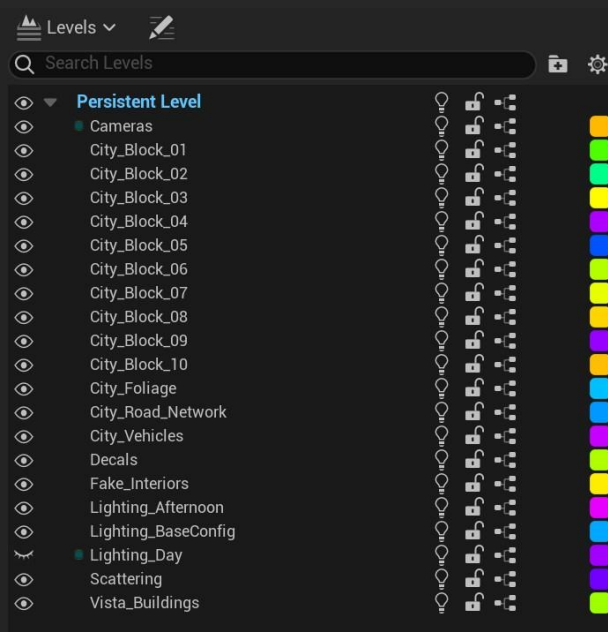


Scene > The main City Level

Overview > Individual assets laid out for preview

Next to this within the main City scene you can also use levels to turn on&off sections of the city. You can do so by going to “Window>Levels”

We divided everything up into blocks. You can also switch between Day/Noon Lighting here



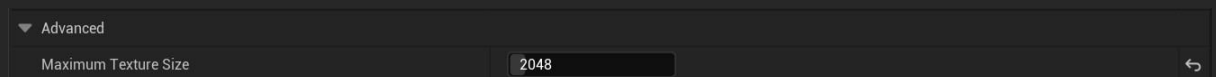
This is pretty much all you need to know to navigate our environment!

Performance

When you download the environment it will come with the highest possible settings. (We did of course run an optimization pass on the lighting, textures, nanite and models)

This might sometimes be too much to handle for older systems. If you experience performance issues, we recommend you do the following:

1. Turn off Raytraced Shadows if on, this is often a big performance impact
2. Set texture resolution lower (We already did this to an extent but when opening a texture you can set a lower resolution by going to Compression > Advanced > Maximum Texture Size.



3. Limit loading in all levels at once in the City but rather use level streaming.

How the environment is constructed.

Structure

The environment Structure is quite unique. In the main level all the buildings and Roads are large static meshes. This is necessary for performance as it is impossible to render hundreds of thousands of assets on a single level. However, all the original roads and buildings were modular and we made sure to include all modular assets in case you want to re-build

sections or build new sections. Foliage is placed using the foliage tool and all other assets are manually placed in their correct streaming level.

We did include interiors in some locations, but these are means to be seen from the outside. Going inside these, the quality will be lower than the exterior.

Assets

Assets have also been made modular where needed. Next to this most of them feature unique textures. (optimized of course)

Some models use unique materials that can change the color of the model based on random position (like the cars)

All assets have nanite on unless they have transparency. They are centered and ready for Drag&Drop

Textures

All textures come in PBR 4k resolution and are after that downsampled in engine to have logical texture resolution and performance. Some of the tiling materials even come in 8k if you desire. However, you will need to open the textures and reset their resolution to use full resolution.

Textures have been packed in the following way:

Color

- **RGB**: Basecolor
- **A**: Cutout alpha (if applicable)

Normal

- **RGB**: Normal Map

ORME

- **R**: Ambient Occlusion
- **G**: Roughness map
- **B**: Metallic
- **E**: Emissive

Please note some textures come un-packed in order to work with world-space shaders

Shaders/Master Materials

Our environment has a core master material. In this material you will have multiple controls

1. Basic controls like tiling, color and intensity controls and of course texture inputs,
2. Detail normal support
3. World Space dust support
4. Color change based on mask support

Next to our master shader we also have some variations based on it which support world space UV mapping, 2-sided rendering and world space color change. (And we have some very basic materials for plain colors, glass, decals, and foliage)

Lighting

We make full use of Lumen in this scene. All assets have been optimized to work with lumen and follow the lumen technical validation documentation (proper thickness, correct distance field and no light errors) – our assets do not come with traditional lightmap UV's but you can always auto generate those if needed.

All lights have been set-up using blueprints with simple on & off + color controls.

Nanite

All of our assets have been set-up using nanite where possible (some assets use traditional LOD's due to transparent materials)

Please note that even though we use nanite. Our assets are created using traditional workflows and are still at standardized game resolution to work in any engine or DCC application. We make heavy use of weighted normal to push the quality of our assets.

Additional Information

1. This project uses **Lumen**, you can learn how to activate it [HERE](#) if needed.
2. All our assets are made while keeping the default UE character scale in mind.
3. Some assets like vehicles and cluster meshes are set up using blueprints for easy placement.
4. Some large-scale assets have been created using the build in modeling tools. You can alter these assets with ease using the same tools.
5. All foliage has been painted using the foliage tool.
6. The focus of this environment was the Exterior, therefore please understand that the interior scenes are less polished.
7. All the branding/Advertisements you see are 100% copyright free and created by our in-house graphic designer.

We hope you enjoyed our city environment!

Disclaimer: This environment is massive and has a lot of moving parts. We did our best to ensure that everything is in working order but if you find any bugs/issues, have questions or concerns please feel free to contact us at: support@scansmatter.com

